How To

Given a rational expression where the factors of the denominator are distinct, irreducible quadratic factors, decompose it.

1. Use variables such as $A, B,$ or C for the constant numerators over linear factors, and linear expressions such as $A_1x + B_1, A_2x + B_2,$ etc., for the numerators of each quadratic factor in the denominator.

$$\frac{P(x)}{Q(x)} = \frac{A}{ax+b} + \frac{A_1x+B_1}{(a_1x^2+b_1x+c_1)} + \frac{A_2x+B_2}{(a_2x^2+b_2x+c_2)} + \cdots + \frac{A_nx+B_n}{(a_nx^2+b_nx+c_n)}$$

2. Multiply both sides of the equation by the common denominator to eliminate fractions.
3. Expand the right side of the equation and collect like terms.
4. Set coefficients of like terms from the left side of the equation equal to those on the right side to create a system of equations to solve for the numerators.